Writing Assignment #8: Complete Induction and Relations

Robert T Kohn

March 31st, 2025

Section 2.5

Question 1

Part B

Claim: $(\forall\ n > 22 \in N)(\exists s, t \in Z)\ \epsilon\ n = 3s + 4t \wedge s \geq 3 \wedge t \geq 2$

Case n=23: s=5, t=2

$3(5) + 4(2) = 15 + 8 = 23$

Case n=24: s=4, t=3

$3(4) + 4(3) = 12 + 12 = 24$

Case n=25: s=3, t=4

$3(3) + 4(4) = 9 + 16 = 25$

Then, $\exists k \in Z \epsilon 23 \leq k \leq n$

Thus, $n \geq 25$

So, $n + 1 = (n - 2) + 3$

Hence, $n - 2 \geq 23 \Rightarrow n + 1 = 3s + 4t + 3$

Then, $n + 1 = 3(s + 1) + 4t$

$\therefore (\forall\ n > 22 \in N)(\exists s, t \in Z)\ \epsilon\ n = 3s + 4t \wedge s \geq 3 \wedge t \geq 2$

Part C

Claim: $(\forall\ n > 33 \in N)(\exists s, t \in Z)\ \epsilon\ n = 4s + 5t \wedge s \geq 3 \wedge t \geq 2$

Case n=34: s=6, t=2

$4(6) + 5(2) = 24 + 10 = 34$

Case n=35: s=5, t=3

$4(5) + 5(3) = 20 + 15 = 35$

Case n=36: s=4, t=4

$4(4) + 5(4) = 16 + 20 = 36$

Case n=37: s=3, t=5

$4(3) + 5(5) = 12 + 25 = 37$

Then, $n + 1 = (n - 4) + 5$

Thus, $n-4 \geq 34 \Rightarrow n+1 = 4s+5t+5$
So, $n+1 = 4s+5t+5 \Rightarrow n+1 = 4s+5(t+1)$
$\therefore (\forall\ n > 33 \in N)(\exists s,t \in Z)\ \epsilon\ n = 4s+5t \wedge s \geq 3 \wedge t \geq 2$

Question 3

Claim: $a_1 = 2 \wedge a_2 = 4 \wedge a_{n+2} = 5a_{n+1} - 6a_n \wedge n \geq 1 \Rightarrow a_n = 2^n$
Case n=1:
$a_1 = 2^1 = 2$
Case n=2:
$a_2 = 2^2 = 4$
Then, for some arbitrary k, $a_k = 2^k$
Thus, $a_{n+1} = 5 * a_n - 6a_{n-1}$
So, $a_{n+1} = 5 * 2^n - 6 * 2^{n-1}$
Hence, $a_{n+1} = 2^{n-1}(5 * 2 - 6 * 1) = 2^{n-1} * 4 = 2^{n-1} * 2^2$
Then, $a_{n+1} = 2^{n+1}$
$\therefore a_1 = 2 \wedge a_2 = 4 \wedge a_{n+2} = 5a_{n+1} - 6a_n \wedge n \geq 1 \Rightarrow a_n = 2^n$

Question 12

Part A

$g1 = 2;$
$g2 = 2;$
$g3 = g2 * g1 = 2 * 2 = 4;$
$g4 = g3 * g2 = 4 * 2 = 8;$
$g5 = g4 * g3 = 8 * 4 = 32;$

Part B

Claim: $g_n = 2^{f_n}$
Case n=1:
$g_1 = 2^{f_1} = 2$
Case n=2:
$g_2 = 2^{f_2} = 2$
Then, for some arbitrary k, $g_i = 2^{f_i}$ for all $i \in \{1, 2...k\}$
Thus, $g_{k+1} = g_k * g_{k-1}$
So, $g_{k+1} = 2^{f_k} * 2^{f_{k-1}} = 2^{f_k + f_{k-1}}$
Hence, $f_{k+1} = f_k + f_{k-1}$
$\therefore g_n = 2^{f_n}$

Section 3.1

Question 6

Part K

$R_{11}^{-1} = \{(x, y) \in P \times P \mid x \text{ is a sibling of } y\}$

Part L

$R_{12}^{-1} = \{(x, y) \in P \times P \mid x \text{ loves } y\}$

Question 7

Part C

$(2, 4) \in S \rightarrow (4, z) \in T \rightarrow (2, 1)$
$(3, 4) \in S \rightarrow (4, z) \in T \rightarrow (3, 1)$
$(3, 1) \in S \rightarrow (3, z) \in T \rightarrow (3, 4)$
$(5, 5) \in S \rightarrow (5, z) \in T \rightarrow (5, z) \notin T$
T $\circ$ S $= \{(2, 1), (3, 1), (3, 4)\}$
I don't know how much work you wanted to see here so I hope this is what you're looking for and let me know if this is too much as well

Part G

$(1, 4) \in T \rightarrow (4, z) \notin S$
$(3, 5) \in T \rightarrow (5, z) \in S \rightarrow (3, 5)$
$(4, 1) \in T \rightarrow (1, z) \notin S$
S $\circ$ T $= \{(3, 5)\}$
$(3, 5) \in$S $\circ$ T $\rightarrow (5, z) \in R \rightarrow (3, 2)$
R $\circ$ (S $\circ$ T)$= \{(3, 2)\}$

Question 11

Part A

Claim: Rng(R^{-1})=Dom(R)
Case 1: Rng(R^{-1}) $\subseteq$ Dom(R)
Suppose $x \in Rng(R^{-1})$,
Then, $\exists y \epsilon (y, x) \in R^{-1}$
Thus, $(x, y) \in R$
So, $x \in Dom(R)$
Case 2: Dom(R) $\subseteq$ Rng(R^{-1})
Suppose $a \in Dom(R)$,
Then, $\exists b \epsilon (a, b) \in R$
Thus, $(b, a) \in R^{-1}$

So, $a \in Rng(R^{-1})$
$\therefore Rng(R^{-1}) = Dom(R)$

Part B

Claim: Dom(S $\circ$R) $\subseteq$ Dom(R)
Suppose x $\notin$ Dom(R)
Then, $\nexists y \epsilon (x, y) \in R$
Thus, $\nexists z (x, z) \in$ S $\circ$R)
Hence, $x \notin Dom$(S $\circ$R)
$\therefore$ Dom(S $\circ$R) $\subseteq$ Dom(R)

Part C

Claim: Dom(S $\circ$R) = Dom(R)
Suppose $A = \{1\}, B = \{2, 3\}, C = \{4\}, S = \{1, 2\}, R = \{3, 4\}$
S $\circ$R=$\emptyset \Rightarrow$ Dom(S $\circ$R)=$\{\emptyset\}$
Dom(R)=$\{1\}$
Dom(S $\circ$R) $\neq$ Dom(R)
$\therefore$ The inequality fails.

Question 14

Claim: $A = m \wedge B = n \Rightarrow 2^{mn}$ different relations from A to B
Case m=1:
Then, $A = \{1\} \wedge B = n$
Thus, there are 2^n possible subsets.
Suppose $A = \{k\} \Rightarrow 2^{kn}$ *relations*
Then, $A = \{k + 1\} \Rightarrow 2^{(k+1)*n}$
$\therefore A = m \wedge B = n \Rightarrow 2^{mn}$ different relations from A to B

Question 17

Part A

This proof gets a grade of F. The claim is false and it assumes that it's composition and inversion commute.

Part B

This proof gets an A.

Part F

This proof gets a grade of A.